Open Source CFD

Tutorial #5: 3D Y-Pipe with FreeCAD

Incompressible Steady Flow: 3D Y-Pipe with simpleFOAM in FreeCAD

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Learning Objectives:

- Set up a complete CFD case using FreeCAD and CfdOF workbench
- Define boundary conditions visually in the GUI
- Generate mesh automatically with snappyHexMesh
- Run simpleFoam directly from FreeCAD
- Post-process results using ParaView integration
- Compare GUI vs terminal workflows (Tutorial #4)

This tutorial shows the work flow for the simulation of a 3D Y-shaped pipe geometry with FreeCAD and the CfdOf workbench. The geometry generated in Tutorial #4 will be reused. Instead of editing manually the dictionaries, in this case all will be done with the GUI. It is more efficient and reduces the risk of typographic mistakes. On the other side, a graphical interface, that often is not available in large computers, is needed.

Contents

1 Opening the Geometry in FreeCAD

1. **Launch FreeCAD** from the terminal or applications menu:
2. **Open the Y-pipe geometry**:
   - Go to `File` → `Open`
   - Navigate to your Tutorial #4 folder
   - Select `Ypipe.FCStd`
   - Click `Open`

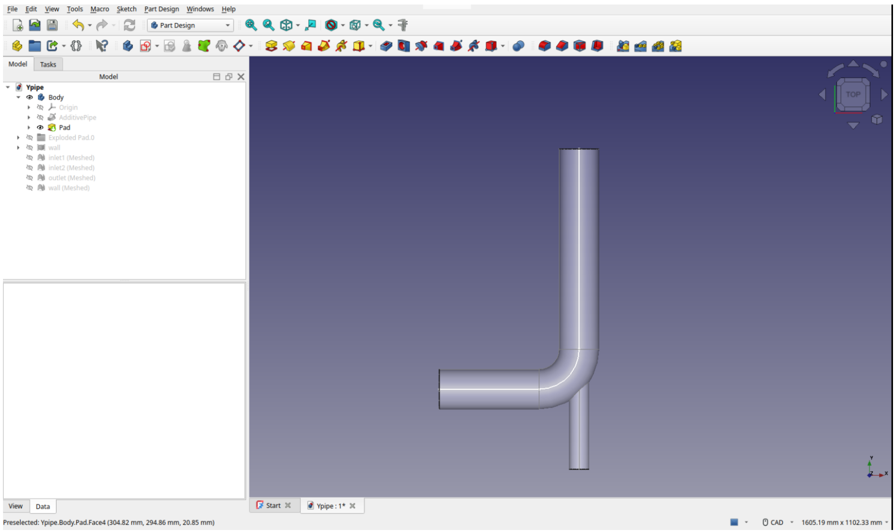

Figure 1: Opening the Y-pipe geometry

3. **Verify the geometry**:
   - The model should appear in the 3D view
   - The tree view should show the `Body` with all features
   - Rotate and zoom to inspect all parts

TIP: If you don't have the geometry file, follow the geometry creation steps from Tutorial #4, Sections 2.1-2.2.

2 Setting Up the CFD Analysis

2.1 Activating CfdOF Workbench

1. **Switch to CfdOF workbench**:
   - From the workbench dropdown, select **CfdOF**
2. **Change output directory**
   - Select the menu `CfdOf` → `Open preferences`
   - Change the `default output directory` to your folder for the Tutorial #5.

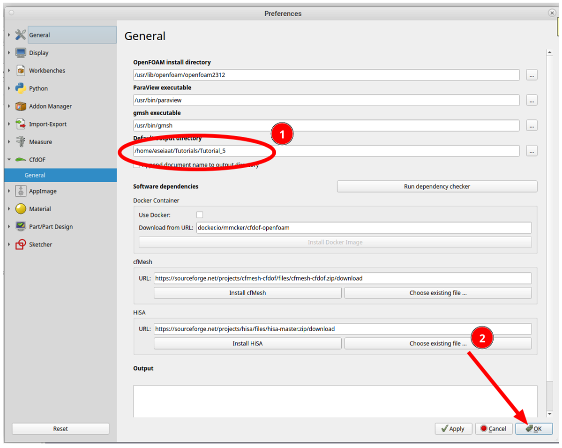

Figure 2: Selecting of output directory

3. **Create a new analysis**:
   - Click the **Analysis container** button in the toolbar
   - A new `CfdAnalysis` object appears in the tree

2.2 Defining the Simulation

1. **Set physics models**:
   - Click on `Select models` button
   - In the Tasks panel, set:
     - `Flow type: Incompressible`
     - `Time: Steady`
     - `Turbulence model: RANS k-omega SST`

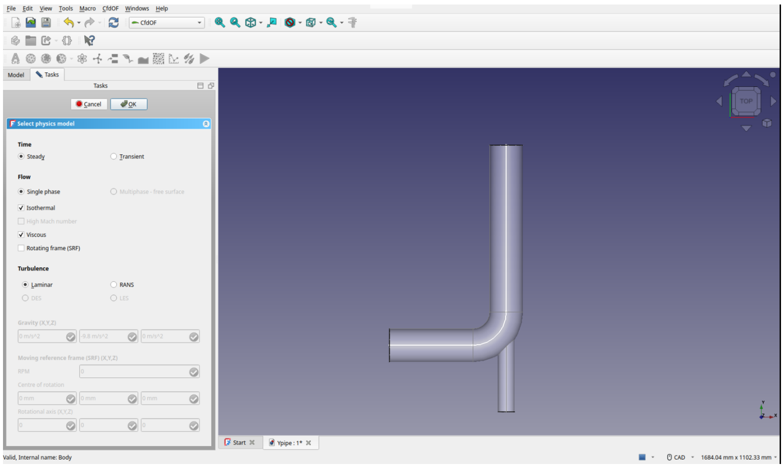

Figure 3: Physics models

2. **Select fluid material**:
    - Click **Add fluid properties** button
    - Choose `Water` from the library
    - Verify properties: $\rho = 1000\,\mathrm{kg/m^3}$, $\nu = 10^{-3}\,\mathrm{Pa \cdot s}$

3 Defining Boundary Conditions

3.1 Identifying Boundary Faces

The Y-pipe has four distinct boundary regions:

- **inlet1**: Secondary inlet (horizontal branch)
- **inlet2**: Primary inlet (vertical pipe)
- **outlet**: Pipe outlet
- **wall**: All remaining surfaces

3.2 Creating Boundary Conditions

1. **Add inlet2**:
    - Select `inlet2` face (primary inlet). It will become green.
    - Click **Fluid boundary** button. It will become blue
    - `Type: Inlet, Uniform velocity`

- `Velocity`: `(0 4 0)` m/s (vertical direction)
- `Pressure`: `0` Pa (relative pressure)
- Click `OK`
- `Turbulence specification: Intensity & Length Scale`
  - `Intensity (I)`: 5%
  - `Length Scale (l)`: 1.4 mm
- Rename it as `inletMain`

TIP: `CfdOf` automatically calculates k and ω from these parameters using standard formulas: $k = \frac{3}{2}(UI)^2$, $\omega = \frac{\sqrt{k}}{C_\mu^{1/4} L}$ where $C_\mu = 0.09$ and L is the mixing length scale.

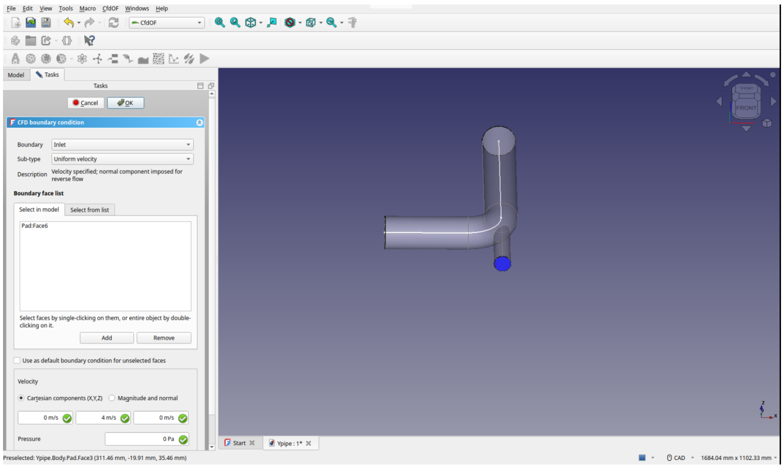

Figure 4: Setting inlet boundary conditions

2. **Add inlet1**:
   - Repeat the same steps as with `inlet2`
   - Type: `inlet, Total pressure`
   - Set `Pressure` : 0 Pa
   - Same turbulence specification as in `inlet2`
   - Rename it as `inletSecondary`
3. **Add outlet**:
   - Repeat the same steps
   - Type: `outlet, Static pressure`

- `Pressure: 0 Pa` (static)
- Leave the default name (`outlet` is not allowed because there is already an element in the tree with this name)

4. **Add walls**:
   - Select all remaining faces (use Ctrl+click for multiple selection)
   - `Type: wall`
   - `Name: walls`
5. **Verify all boundaries**:
   - The tree should show four boundary conditions
   - Each should have a distinct color in the 3D view

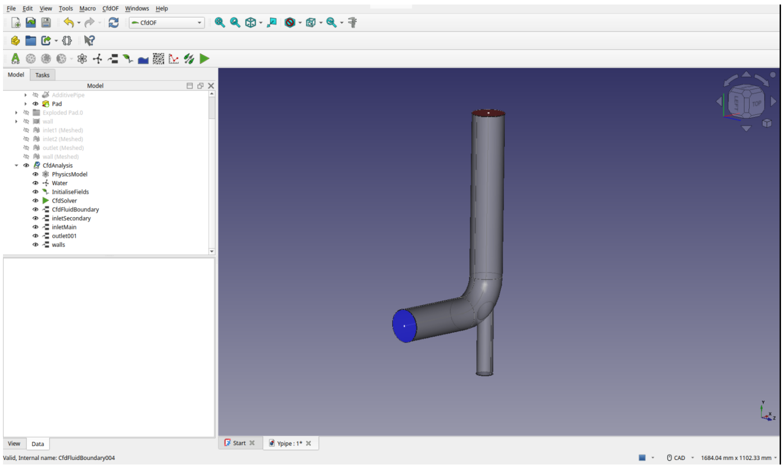

Figure 5: Completed boundary conditions

4 Mesh generation with `snappyHexMesh`

- With the `Pad` selected, click on `CFD Mesh`
- Choose `snappyHexMesh` as mesh utility
  - Define the `Base element size` as 5 mm
  - Clik on `Search` to pick an inner point for the mesh
  - Leave the other parameters with the default values
  - Click on `Write mesh case` to write the case for meshing in the disk. Optionally the case can be reviewed

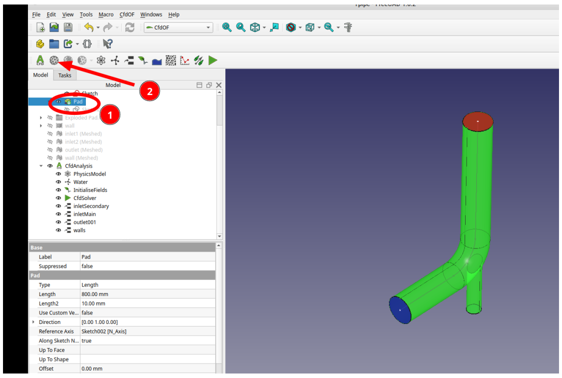

Figure 6: Adding a mesh to the model

  - Click on `Run mesher`
  - When the mesh is complete it can be viewed with `ParaView`. Teh result should be similar to the `snappy` stage of Tutorial #4.

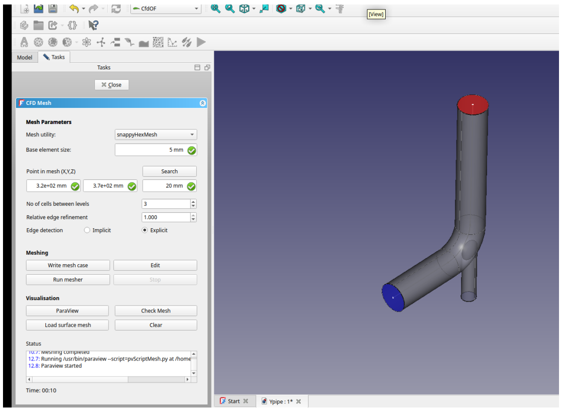

Figure 7: Setup of mesh with `snappyHexMesh`.

- To create the boundary layer, select the mesh element and click on `Mesh refinement` (see Figure 8)
  - Leave the default selection `Surface refinement`
  - Write 1.0 as `Relative mesh size`.
  - Click on the `Boundary layers` check box.
  - `Number of layers` → `3`
  - `Expansion ratio` → `1.2`
  - Select the `wall` element from the list (second tab on the last box). Optionally it can be selected in the model, but it is easier to select from the list.

– Save with `Ok`

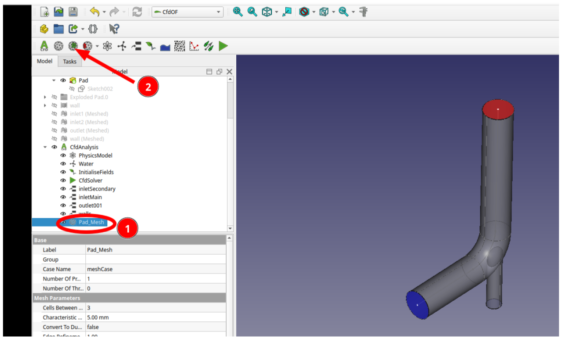

Figure 8: Add the boundary layer to the mesh

- Go back to the `Pad_Mesh` element, write again the case mesh and run the mesher to get the mesh with the boundary layer. It should be similar to the mesh generated in Tutorial #4.

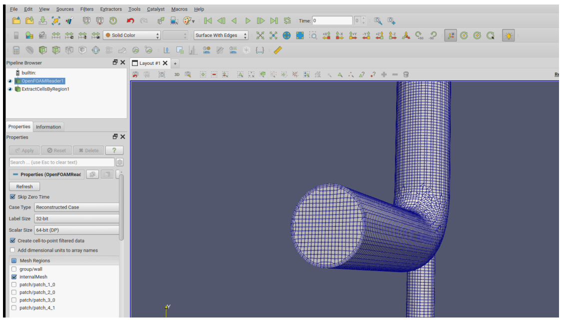

Figure 9: Final mesh viewed in `ParaView`

5 Initialisation of the case and running the case

– Click on `Initialise` to get the initial fields for the simulation
  - ∗ Check `Velocity` → `Potential flow`
  - ∗ Check `Pressure` → `Potential flow`
  - ∗ Check `Turbulence` → `Use value from boundary` → `inletSecondary`
  - ∗ Save with `OK`

– Select the element `CfdSolver` and change the `Parallel` parameter to `false`

– Change the maximum number of iterations to 1000

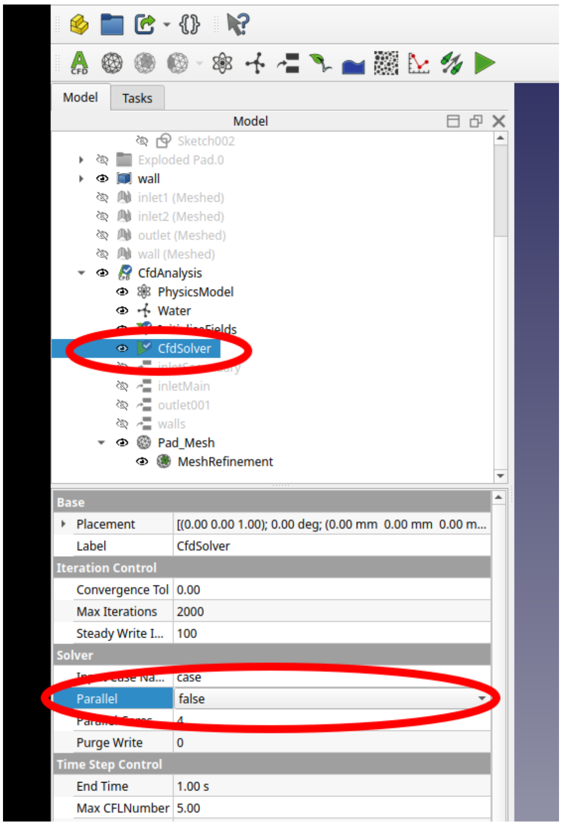

Figure 10: Disable the parallel simulation

– Double-click on `CfdSolver` and write the case.

– Run the solver. It will not converge completely. We should change numerical schemes or other solver settings in order to get a better convergence, but it is beyond the scope of this Tutorial. It will take about 5 minuts to complete the 100 iterations

6 Visualisation of computation of flow rate with `ParaView`

* Open the results with `ParaView` from the solver box
* Visualisation of fields is similar to the description in Tutorial #4.
* To compute flow rate in the `inletSecondary` patch, select uniquely this in the `OpenFOAMReader1` element
* In the canvas, only the `inletSecondary` patch will be visible.
* Apply the filter `Integrate Variables`, with `Cell data as atribute`.

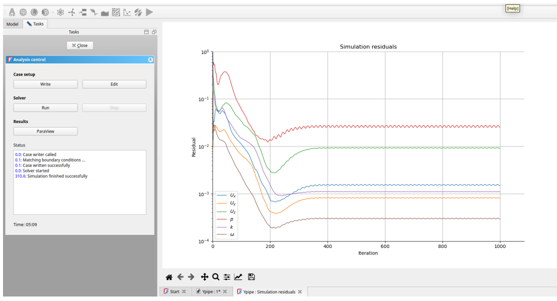

Figure 11: Residuals of the simulation

* The flow rate is the integral of the component x of the velocity U (see Figure 13)

7 Exercises

The same exercises of Tutorial #4 can be done here.

7.1 Question about flow rate

The flow rate obtained in this tutorial is different from the one computed in Tutorial #4. Can you see why? How should the simulation be changed in order to get the same flow rate?

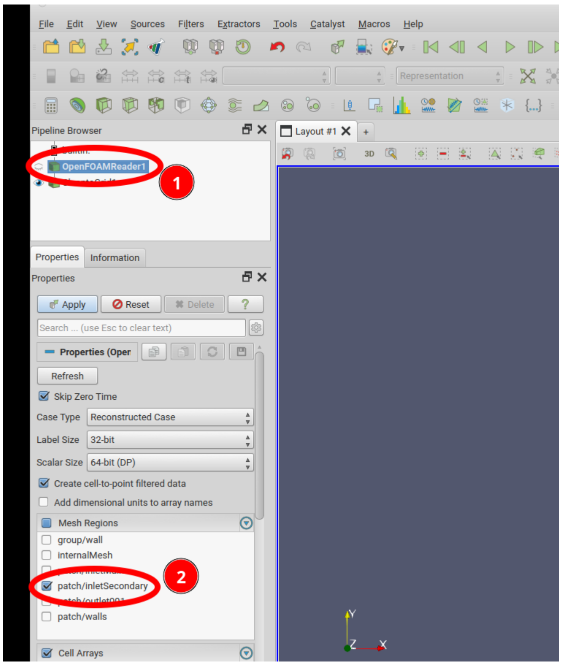

Figure 12: Selection of `inletSecondary` patch in the visualisation.

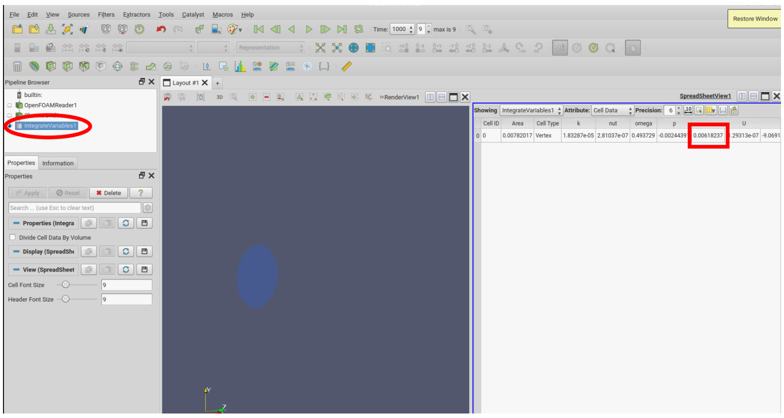

Figure 13: Computation of flow rate in `inletSecondary`